ESE 507

Project: Part 1

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1. Multiply Accumulate (MAC) Design

Two versions of the MAC were implemented to compare the unpipelined and pipelined designs. Both the unpipelined `mac` and pipelined `mac_pipe` designs use an accumulator register that updates its output on the rising edge of the clock. The primary difference between the two designs is the combinational multiply-add operation. In the unpipelined `mac`, the multiplication and addition are performed within a single combinational stage before the result is stored in the accumulator. In the pipelined `mac_pipe`, the multiply-add operation is divided into two stages. The multiplier output and associated control signals are stored in an additional intermediate register before the addition is performed and the result is stored in the accumulator on the following clock cycle. Consequently, the pipelined design introduces an additional cycle of latency, requiring two clock cycles for an input to propagate to the accumulator output, compared with one cycle for the unpipelined design. However, the benefits of pipelining become apparent when considering the overall throughput and frequency of the system.

1.1. Design Parameter

Both the `mac` and `mac_pipe` designs were synthesized with `WIDTH = 16` and `ACCW = 48` across a range of clock frequencies. The analysis spans frequencies from a relatively low frequency of 10 MHz to a maximum of 1 GHz. At each frequency, the area, power, critical path location, and timing constraint of each design are evaluated.

2. Combinational Synthesis Results

The unpipelined `mac` design was synthesized to determine the maximum achievable clock frequency with `WIDTH = 16` and `ACCW = 48`. The synthesis workflow was automated with scripts that write the configured design parameters and the synthesis script, before running synthesis on the remote CAD server. The generated synthesis report is filtered for key attributes, including area, power, and timing. For convenient data analysis, these attributes are stored in comma-separated value (CSV) files for every synthesis run. The tabular structure of the CSV files is well suited for Python plotting libraries, which were used to generate the graphs below. For more details on the automation workflow, refer to `scr/`. Only data points that **meet the timing constraint** are plotted in the graphs below.

2.1. Area

The design area is estimated from the synthesis report as the total cell area, which does not account for net interconnect area. Therefore, the actual physical area of the design with interconnects may be larger than the reported area. Further analysis of design area uses the conservative estimation as the sum of combinational area, buffer/inverter area, noncombinational area and macro/black box area in the synthesis report.

As shown in Figure 1, the total design area remains relatively constant as the frequency increases. However, an overall upward trend is seen, particularly at frequencies above 200 MHz. This increase can be attributed to the additional area required to meet the more difficult timing constraints associated with higher frequency, where the trade-off for frequency is at the expensive of area. In contrast, the relatively constant area at lower frequencies suggests that the synthesis tool implemented the design to the near-minimal optimized cell configuration regardless of less restrictive timing constraints.

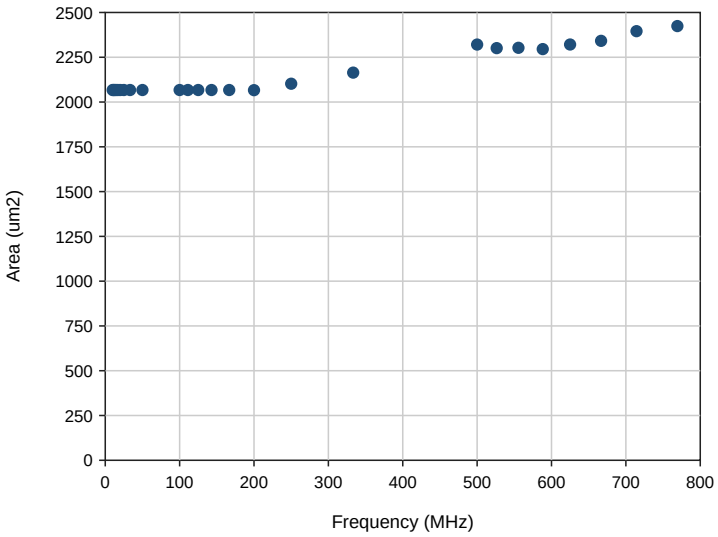


Figure 1: Area, frequency for unpipelined `mac` (MET).

Freq. (MHz)	Area (μm^2)
10	2067.09
20	2067.09
25	2067.09
50	2067.09
100	2067.09
166.67	2067.09
200	2066.29
333.33	2164.18
⋮	⋮
500	2320.85
526.32	2300.90
555.56	2302.76
769.23	2424.06

Table 1: Selected MET area points for `mac`.

An interesting pattern is the occasional small downward trend in area at high frequencies, between 500 MHz and 769.23 MHz. At these frequencies, the synthesis tool may had identified a simplified version that meets timing constraint with less cell area. For example, the design synthesized at 555.56 MHz has a slightly lower total area than the design synthesized at 500 MHz, despite the overall increasing trend of frequency and area. This behavior indicates that the relationship between frequency and area is not strictly one dimensional and can depend on the specific optimizations selected by the synthesis tool.

2.2. Power

The Total Power shown in Figure 2 is defined as the sum of Total Dynamic Power and Cell Leakage Power in the synthesis report. The Total Dynamic Power is also reported as the sum of Cell Internal Power and Net Switching Power, both contributing an near equal portion of the sum.

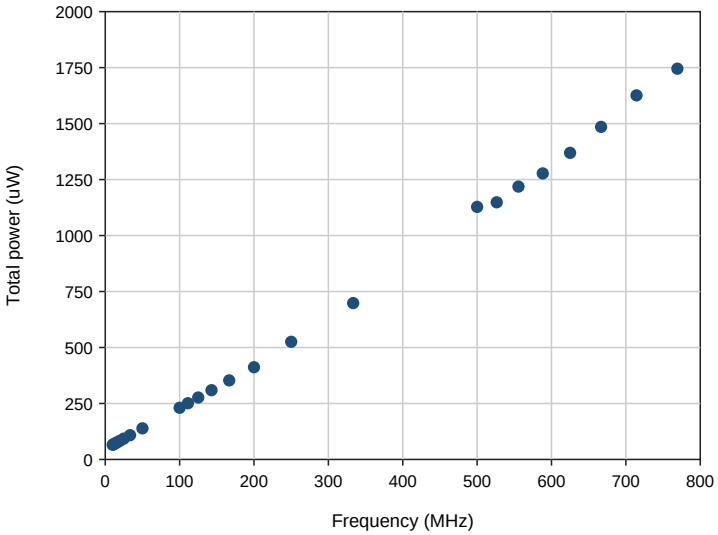


Figure 2: Total power, frequency for unpipelined `mac` (MET).

Freq. (MHz)	$\sum$ Power (μW)
10	65.49
20	83.85
25	93.03
50	138.92
100	230.72
166.67	353.12
200	411.86
333.33	698.23
⋮	⋮
500	1127.93
526.32	1148.45
555.56	1218.61
769.23	1745.19

Table 2: Selected MET power points for `mac`.

The Cell Leakage Power is almost negligible compared with the dynamic power. This may be attributed to the relatively small area of the design, with the accumulator register being the primary memory elements contributing to leakage power. In the case that the frequency is considerably slow, the dynamic power is sufficiently small that leakage power becomes a significant contributor to total power. Otherwise, given a fixed design area, the leakage power remains approximately constant at frequency below 166.67 MHz, as shown in the table of Figure 1. This suggests that leakage power is primarily dependent on the area of the design. The slight increase in leakage power at higher frequencies can therefore be attributed to the corresponding increase in design area.

Consequently, the trend seen in Figure 2 for total power is largely influenced by the Dynamic Power and its dependence on frequency. Increasing the frequency results in a near linear increase in Dynamic Power, with an approximate power increase of 1.836 μW/MHz. In other words, a higher-frequency design, performing more switching operations, will consume more power over the shorter period of time.

2.3. Critical Paths

The synthesis report provides the start and end points of the critical paths, along with the paths themselves and their data arrival times. If the data arrival time is less than or equal to the data required time, the design constraint is considered MET, as indicated by Slack (MET) in the report.

A design with Slack (VIOLATED) indicates a timing constraint violation and will typically display a negative slack value, calculated as the difference between the data arrival time and the data required time. A negative value indicates that the timing constraint has not been met. A design reported as Slack (VIOLATED: increase significant digits) is also considered a violation, even if the displayed slack value is zero. This occurs because the actual slack value is slightly negative but is rounded to zero due to the limited number of significant digits displayed in the report.

Freq. (MHz)	Area (μm^2)	Power (μW)	Slack (ns)	Path Start	Path End	Timing
10	2067.09	65.49	95.68	input0[1]	acc_reg[47]	MET
11.11	2067.09	67.53	85.68	input0[1]	acc_reg[47]	MET
12.5	2067.09	70.08	75.68	input0[1]	acc_reg[47]	MET
14.29	2067.09	73.35	65.68	input0[1]	acc_reg[47]	MET
16.67	2067.09	77.73	55.68	input0[1]	acc_reg[47]	MET
20	2067.09	83.85	45.68	input0[1]	acc_reg[47]	MET
25	2067.09	93.03	35.68	input0[1]	acc_reg[47]	MET
33.33	2067.09	108.32	25.68	input0[1]	acc_reg[47]	MET
50	2067.09	138.92	15.68	input0[1]	acc_reg[47]	MET
100	2067.09	230.72	5.68	input0[1]	acc_reg[47]	MET
111.11	2067.09	251.12	4.68	input0[1]	acc_reg[47]	MET
125	2067.09	276.62	3.68	input0[1]	acc_reg[47]	MET
142.86	2067.09	309.40	2.68	input0[1]	acc_reg[47]	MET
166.67	2067.09	353.12	1.68	input0[1]	acc_reg[47]	MET
200	2066.29	411.86	0.68	input0[1]	acc_reg[47]	MET
250	2102.20	525.46	0.01	input0[0]	acc_reg[47]	MET
333.33	2164.18	698.23	0.00	input0[7]	acc_reg[47]	MET
500	2320.85	1127.93	0.00	input0[3]	acc_reg[47]	MET
526.32	2300.90	1148.45	0.03	input0[5]	acc_reg[47]	MET
555.56	2302.76	1218.61	0.00	input0[7]	acc_reg[47]	MET
588.24	2295.58	1277.55	0.01	input0[11]	acc_reg[47]	MET
625	2321.12	1369.41	0.00	input0[5]	acc_reg[47]	MET
666.67	2341.60	1485.30	0.00	input0[5]	acc_reg[44]	MET
714.29	2395.60	1626.06	0.00	input0[0]	acc_reg[35]	MET
769.23	2424.06	1745.19	0.00	input0[7]	acc_reg[31]	MET
833.33	2471.14	2003.27	−0.07	input0[5]	acc_reg[33]	VIOLATED
909.09	2543.23	2229.30	−0.14	input0[15]	acc_reg[21]	VIOLATED
1000	2529.13	2441.94	−0.22	input0[7]	acc_reg[38]	VIOLATED

Table 3: Frequency, area, power, timing slack, and critical path for unpipelined `mac`.

The critical path starts from an `input0` register and ends at `acc_reg`, with slight variations in the indices of `input0` and `acc_reg`. At low frequencies and up to approximately 200 MHz, the critical path remains the same, starting from `input0[1]` and ending at `acc_reg[47]`. Between 200 MHz and 625 MHz, the endpoint of the critical path remains fixed at `acc_reg[47]`, while the index of `input0` varies depending on the frequency. Above 625 MHz, both the `input0` index and the endpoint index of `acc_reg` vary without a clear or consistent pattern as shown in Table 3.

In particular, there appears to be a tendency for the critical path to shift toward higher-indexed `input0` elements and, in other cases, lower-indexed `acc_reg` elements. This indicate a possible trend in which the synthesis fail to optimize the design of higher indexed paths, in contrast to the critical path at lower frequency to be at lower indexed of input0. Another explination is that the paths between different indexed elements have the same critical path or delay, which synthesis does not differentiate. Therefore, additional analysis is required to determine whether this trend is directly caused by the stricter timing constraints, the design area, or their interaction.

3. Pipelined Synthesis Results

In a similar fashion, the pipelined `mac_pipe` design was synthesized to determine the maximum achievable clock frequency with `WIDTH = 16` and `ACCW = 48`. The same automated scripts were used to collect and store the synthesis results in a CSV file. The resulting datasets for the unpipelined `mac` and pipelined `mac_pipe` designs were then analyzed to compare their area, power, and timing performance. In terms of performance metric, the maximum achievable frequency was prioritized before considering tradeoff in area and power, whose relative importance depends on the target application. Only data points that meet the timing constraint are plotted in the graphs below.

3.1. Area

The design area of the pipelined mac_pipe is the estimated cell area from the synthesis report. For better comparison, the area for both the unpipelined and pipelined designs are plotted over frequency.

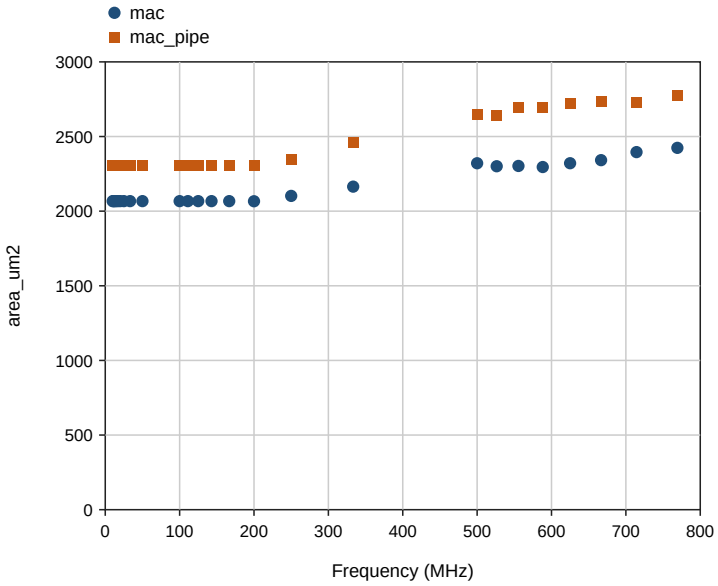


Figure 3: Area between `mac` and `mac_pipe` (MET).

Freq. (MHz)	Area (μm^2)		ΔArea
	`mac`	`mac_pipe`	
10	2067.09	2306.75	239.67
20	2067.09	2306.49	239.40
25	2067.09	2306.49	239.40
50	2067.09	2306.49	239.40
100	2067.09	2306.49	239.40
166.67	2067.09	2306.49	239.40
200	2066.29	2305.16	238.87
333.33	2164.18	2459.44	295.26
⋮	⋮	⋮	⋮
500	2320.85	2648.03	327.18
526.32	2300.90	2639.78	338.88
555.56	2302.76	2692.98	390.22
769.23	2424.06	2773.58	349.52

Table 4: Selected MET area points.

As shown in Figure 3, the area of the pipelined design is generally higher than the unpipelined design, which is expected given the additional pipeline registers. Accounting for increasing in frequency, the area of the pipelined design is about 240 μm^2 higher than the unpipelined design at 10 MHz, increasing to about 320 μm^2 at 500 MHz. This trend suggests that the pipelined design is area-deficient at higher frequencies.

An interesting trend in area occurs at the particular frequency from 500MHz to 700MHz. At those frequency the area of the pipelined design increase while the unpipelined design decrease, such that the area difference is significantly greater than at other frequencies. This suggests that pipelined design is more area-deficient at frequencies where the unpipelined design could be optimized to be more area-efficient.

3.2. Power

The Total Power is defined the same way as the unpipelined design. Previously, dynamic power was analyzed to cause the trend seen in Figure 2. This trend is also seen in Figure 4 for the pipelined design.

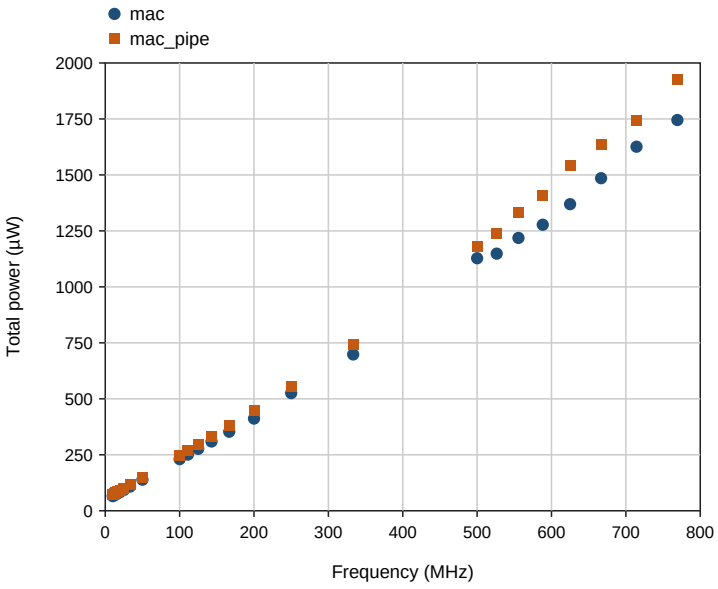


Figure 4: Power between `mac` and `mac_pipe` (MET).

Freq. (MHz)	∑ Power (μW)		ΔPower
	`mac`	`mac_pipe`	
10	65.49	71.03	5.55
20	83.85	90.75	6.90
25	93.03	100.61	7.59
50	138.92	149.92	10.99
100	230.72	248.53	17.81
166.67	353.12	380.01	26.89
200	411.86	446.49	34.63
333.33	698.23	744.59	46.36
⋮	⋮	⋮	⋮
500	1127.93	1182.18	54.26
526.32	1148.45	1239.58	91.13
555.56	1218.61	1333.95	115.34
769.23	1745.19	1928.30	183.11

Table 5: Selected MET power points.

Cell Leakage Power continues to be almost negligible compared with dynamic power, as in the unpipelined design. The slight increase in leakage power for the pipelined design is attributed to the additional pipeline registers increasing the cell area over which leakage can occur. Thus the leakage power could be a more significant fraction in larger designs, especially at lower frequencies. Otherwise, leakage stays approximately constant at low frequency as seen in unpipelined design with the same parameters. The slight increase in power at higher frequency can be attributed to the increased design area from synthesis.

The power difference is minimal at lower frequency, but growing at a larger rate than the unpipelined design. This can be a consequence of the larger pipelined area providing more cells that will be switched each cycle. The rate of power growth becomes more distinct at higher frequency. For example, ΔPower remains on the order of tens of microwatts through 500 MHz and then continues to peak, reaching about 183 μW at 769.23 MHz. At this rate, the pipelined design may significantly exceed the power budget compared to the unpipelined design at increasing frequency.

3.3. Critical Paths

The start and end points of the critical path for `mac_pipe` are reported in Table 6. Because the multiply-add operation is split across two registers, the critical path may lie in either the multiplier stage `reg1` or the accumulator stage `reg2`.

Freq. (MHz)	Area (μm^2)	Power (μW)	Slack (ns)	Path Start	Path End	Timing
10	2306.75	71.03	95.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
11.11	2306.75	73.22	85.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
12.5	2306.49	75.96	75.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
14.29	2306.75	79.48	65.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
16.67	2306.75	84.18	55.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
20	2306.49	90.75	45.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
25	2306.49	100.61	35.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
33.33	2306.49	117.05	25.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
50	2306.49	149.92	15.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
100	2306.49	248.53	5.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
111.11	2306.49	270.44	4.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
125	2306.49	297.83	3.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
142.86	2306.49	333.05	2.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
166.67	2306.49	380.01	1.74	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
200	2305.16	446.49	0.75	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
250	2342.93	554.37	0.12	reg1/out_product_reg[0]	reg2/out_accum_reg[47]	MET
333.33	2459.44	744.59	0.00	reg2/out_accum_reg[0]	reg2/out_accum_reg[47]	MET
500	2648.03	1182.18	0.00	reg2/out_accum_reg[9]	reg2/out_accum_reg[47]	MET
526.32	2639.78	1239.58	0.00	reg2/out_accum_reg[7]	reg2/out_accum_reg[46]	MET
555.56	2692.98	1333.95	0.00	input0[7]	reg1/out_product_reg[30]	MET
588.24	2693.52	1407.20	0.01	input0[3]	reg1/out_product_reg[30]	MET
625	2722.78	1541.62	0.00	input0[5]	reg1/out_product_reg[30]	MET
666.67	2732.35	1636.92	0.00	input0[5]	reg1/out_product_reg[29]	MET
714.29	2724.64	1741.75	0.00	input0[3]	reg1/out_product_reg[30]	MET
769.23	2773.58	1928.30	0.00	reg2/out_accum_reg[7]	reg2/out_accum_reg[47]	MET
833.33	2823.86	2116.52	0.00	input0[11]	reg1/out_product_reg[23]	VIOLATED[1]
909.09	2860.56	2372.62	−0.06	input0[13]	reg1/out_product_reg[27]	VIOLATED
1000	2832.37	2581.64	−0.18	input0[9]	reg1/out_product_reg[20]	VIOLATED

Table 6: Frequency, area, power, timing slack, and critical path for pipelined `mac_pipe`.

The additional pipeline register splits the multiply-add operation into two stages, so the critical path may lie in either the multiplier stage or the accumulator stage. At low frequencies and up to around 200 MHz, the critical path is consistently from index [0] to index [47] of `out_accum_reg` in `reg2`. This is likely the path of 48-bit adder carry chain from the least-significant bit of the accumulator to the most-significant bit. At 250 MHz, the cirtical path indictate the carry chain start from the input of the product rather than the accumulator. From 333.33 MHz through 526.32 MHz the path remains in the accumulate stage, with some variation in the start and end indices of `out_accum_reg`. This suggests the carry chain is the main bottleneck at lower frequencies.

[1]The 833.33 MHz point has a displayed slack of zero but considered as VIOLATED because of negative slack being rounded to zero.

Between 555.56 MHz and 714.29 MHz the critical path moves to the multiplier stage, starting from an `input0` index and ending at `reg1/out_product_reg`. This indicates the 16×16 multiplier is now the critical path at these frequencies. At 769.23 MHz, the highest frequency that meets timing, the path returns to the accumulate stage. Timing violations from 833.33 MHz onward are again on the multiplier path, from `input0` to `reg1/out_product_reg`. This suggests the multiplier stage is the bottleneck preventing the design from achieving higher frequency.

4. Performance Comparison

Both the pipelined and unpipelined design MET the timing constraint at a maximum frequency of 769.23 MHz, corresponding to a period of 1.30 ns. However, the slack violation due to negative rounding indicates that the pipelined design was close to meeting the timing constraint at a period of 1.20 ns. A greater resolution of the clock period may capture a faster pipelined design.

Design	Period (ns)	Freq. (MHz)	Area (μm^2)	Power (μW)	Slack (ns)	Path Start	Path End	Timing
mac	1.29	775.19	2388.68	1752.89	−0.06	input0[7]	acc_reg[36]	VIOLATED
mac	1.28	781.25	2401.98	1785.59	−0.05	input0[3]	acc_reg[26]	VIOLATED
mac	1.27	787.40	2415.01	1802.77	−0.04	input0[6]	acc_reg[38]	VIOLATED
mac	1.26	793.65	2462.89	1893.45	−0.04	input0[9]	acc_reg[39]	VIOLATED
mac	1.25	800.00	2445.34	1855.99	−0.05	input0[8]	acc_reg[43]	VIOLATED
mac	1.24	806.45	2485.50	1892.75	−0.05	input0[5]	acc_reg[41]	VIOLATED
mac	1.23	813.01	2469.28	1908.01	−0.06	input0[13]	acc_reg[42]	VIOLATED
mac	1.22	819.67	2433.63	1872.99	−0.07	input0[1]	acc_reg[42]	VIOLATED
mac	1.21	826.45	2441.61	1933.98	−0.04	input0[6]	acc_reg[23]	VIOLATED
mac_pipe	1.21	826.45	2810.29	2122.06	0.00	input0[8]	reg1/out_product_reg[25]	MET

Table 7: Fine-period area, power, slack, and timing for maximum frequency.

As shown in Table 7, the pipelined design is able to meet a stricter timing constraint with a clock period of 1.21 ns. Where the unpipelined design is VIOLATED at 1.21 ns, the pipelined design MET the timing constraints to run at a faster frequency of 826.45 MHz. Therefore the additional pipeline register provide a modest 60MHz improvement in frequency compared to the unpipelined design. The critical path is still from `input0` to `reg1/out_product_reg` at 826.45 MHz, indicating the multiplier stage is still the bottleneck of the pipelined design. Achievement in higher frequency may be possible with deeper pipelining of the multiplier stage.

For now, we will use the pipelined design at a clock period of 1.21 ns or 826.45 MHz as the maximum clock frequency for the following analysis.

4.1. Maximum-Frequency Energy

Energy is the integral of power over time. Synthesis reports average power,

$$P_{\text{avg}} = \frac{1}{T}\int_0^T P(t)\,dt,$$

in Watts as Joules per second. Using the maximum frequency case from Table 7, P_{avg} is the total power of the `mac_pipe` design. The time is the clock period used that MET the timing constraint for `mac_pipe`, where the clock period is $t = 1.21$ ns for the maximum frequency case. Therefore, the energy of a single clock cycle is

$$E_{\text{cycle}} = P_{\text{avg}}\,t.$$

Suppose we want to compute 50 sets of input values. The 2-stage `mac_pipe` design will take one additional cycle to fill the pipeline with inputs before the outputs could be computed every cycle. Then the total time to compute 50 sets will require 51 cycles. Thus, the energy consumed for 50 sets of input is

$$\begin{aligned} E = 51(E_{\text{cycle}}) &= 51(P_{\text{avg}}\, t) \\ &= 51(2.122 \times 10^{-3}\ \text{W} \times 1.21 \times 10^{-9}\ \text{s}) \\ &= 1.31 \times 10^{-10}\ \text{J}. \end{aligned}$$

This is equivalent to about 131 pico-Joules of energy consumed to compute 50 sets of input values.

4.2. Frequency-Dependent Energy

The energy computed for 50 input sets at the maximum frequency would change if the same `mac_pipe` design were resynthesized at a different target clock. At every MET point the energy is still

$$E = P_{\text{avg}}\, t = \frac{P_{\text{avg}}}{f},$$

so a change in the target period changes both t and the synthesized P_{avg}. Figure 5 plots the product $P_{\text{avg}}\, t$ against frequency. The product is the energy of one cycle, multiplying by 51 cycles gives the energy to compute 50 input sets.

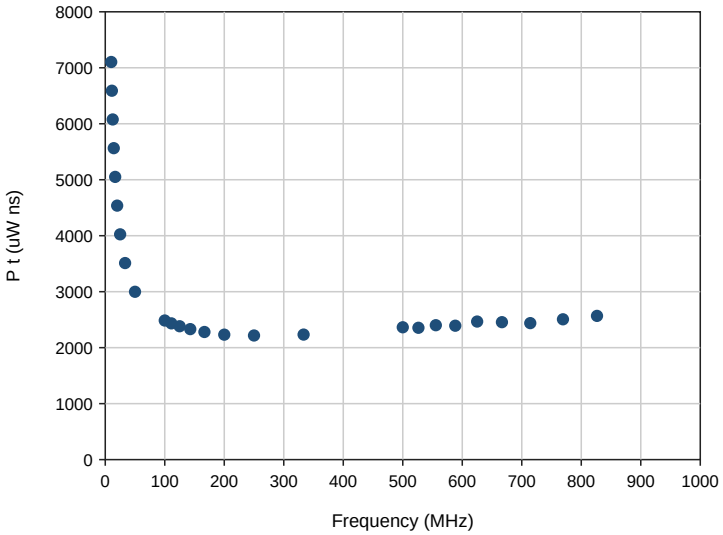


Figure 5: Energy per cycle $P_{\text{avg}}\, t$ for `mac_pipe`.

Freq. (MHz)	E **(pJ)**
10	362.27
20	231.41
25	205.25
50	152.92
100	126.75
166.67	116.28
200	113.85
333.33	113.92
⋮	⋮
500	120.58
526.32	120.12
555.56	122.46
769.23	127.85
826.45	130.95

Table 8: Frequency on Energy per cycle.

The Energy trend as the integeral of the linear power plot in Figure 5 is not entirly constant. At very low frequency, where the constant leakage power is dominant, the energy cost is drastically high. This is from increasing period time that the leakage power does not compenstate, raising the energy product equation. Only at higher frequency of around 100 MHz and above, the energy cost is approximate constant from the linear relationship between dynamic power and frequency. If we just account for the dynamic power, the energy cost would be approximately constant with increasing frequency. The slight linear trend in energy can be attributed to the slight increase in area, contributing to total power at higher frequencies.

4.3. Latency and Throughput

Latency is the time it takes for one unit input to be computed. This is the total time by the number of cycles an input must pass through the design. In case of the pipelined design, each input take two cycles to pass through two registers, resulting in a latency of two periods. The unpipelined design takes one cycle to for the input topass through the design, resulting in a latency of one period. Then latency L is

$$L = N_{\text{cyc}}\, t,$$

where N_{cyc} is the number of cycles and t is the period for every cycle.

Throughput is the number of outputs computed per unit time. If we disregard the fill time or the latency of the first input in the pipeline design, both the pipelined and unpipelined design will produce one output per cycle. Then throughput F is just the inverse of the period or the frequency of the design given by

$$F = \frac{1}{t},$$

Where F can also be interpreted as the amount of MAC operations per second.

Table 9 shows the latency and throughput, defined above, along with the area and power of the pipelined and unpipelined design at the maximum MET timing constraint.

Design	**Area (μm^2)**	**Power (μW)**	**Period (ns)**	N_{cyc}	**Latency (ns)**	**Throughput (MAC/s)**
`mac`	2424.06	1745.19	1.30	1	1.30	7.69×10^8
`mac_pipe`	2810.29	2122.06	1.21	2	2.42	8.26×10^8
Δ (%)	+15.9	+21.6	−6.9	+100	+86.2	+7.4

Table 9: Area, power, latency, and throughput at the maximum MET frequency.

Compared to the unpipelined design, the pipelined design has a higher latency due to the additional pipeline cycle, but greater throughput by meeting the timing constraint at a higher frequency.

4.4. Design Trade-offs

From Table 9, neither design is inherently better than the other. There are trade-offs between area, power, latency, and throughput and depening on the application, one design may be better suited for the task.

A designer should choose `mac_pipe` when the application is limited by throughput or clock rate and can afford the extra area and power. That is the case for a long stream of MACs, where the one-cycle fill is paid once and then every later result uses the faster 1.21 ns period. The pipelined design is also the only option if higher frequency is the main performance metric.

The unpipelined `mac` is the better choice when area, power, or single-result latency matters more than the small throughput increase. A 7.4% gain in throughput may not justify 15.9% more area and 21.6% more power in a tight budget design and that it does not run continuously to utilize the extra throughput.

5. Potential for Deeper Pipelining

5.1. Multiplier Pipelining

5.2. Adder Pipelining

Pipelining the adder is another possible way to potentially improve the frequency of the design. However, it is not necessary if bottleneck is in the multiplier stage, attempts to pipeline the adder would not improve the critical path. Instead, it would only add more area and power to the design. Pipelining the adder would only be a good idea if the critical path is in the accumulator stage.

6. Variable Parameter Synthesis Results

6.1. Accumulator Width (ACCW)

6.2. Numerical-Bit Width (WIDTH)

6.3. Scaling Behavior

6.4. Accumulator Overflow

7. Appendix